Assignment 9: Centralized Logging Solution

Objective: Implement centralized log management using Loki, Promtail, and Grafana.

GitHub Repository

Repository Name: assignment9-logging

Repository Structure

assignment9-logging/
■■■ docker-compose.yml
■■■ loki/
■ ■■■ local-config.yaml
■■■ promtail/
■ ■■■ config.yml
■■■ grafana/
■ ■■■ dashboards/
■■■ screenshots/
■■■ docs/
■ ■■■ architecture.md
■ ■■■ troubleshooting-guide.md
■ ■■■ rollback-guide.md
■■■ README.md

Objective and Solution Overview

A centralized logging solution was implemented using: • Loki for log aggregation and storage • Promtail for log collection and forwarding • Grafana for visualization and analysis The solution collects application logs and Nginx logs from Docker containers and provides centralized search and monitoring capabilities.

Implementation Steps

1. Created project directory structure. 2. Configured Loki as centralized log storage. 3. Configured Promtail to collect Docker and Nginx logs. 4. Deployed Loki, Promtail, Grafana, and sample Nginx application using Docker Compose. 5. Connected Grafana to Loki data source. 6. Created dashboards for log visualization. 7. Implemented log filtering and search queries. 8. Verified log collection from application and Nginx containers.

Logging Architecture

Nginx Application Logs ↓ Promtail ↓ Loki ↓ Grafana ↓ Log Search, Filtering & Dashboards

Docker Compose Services

• Loki • Promtail • Grafana • Nginx Log Demo Application Ports: - Grafana : 3001 - Loki : 3100 - Nginx Demo : 8086

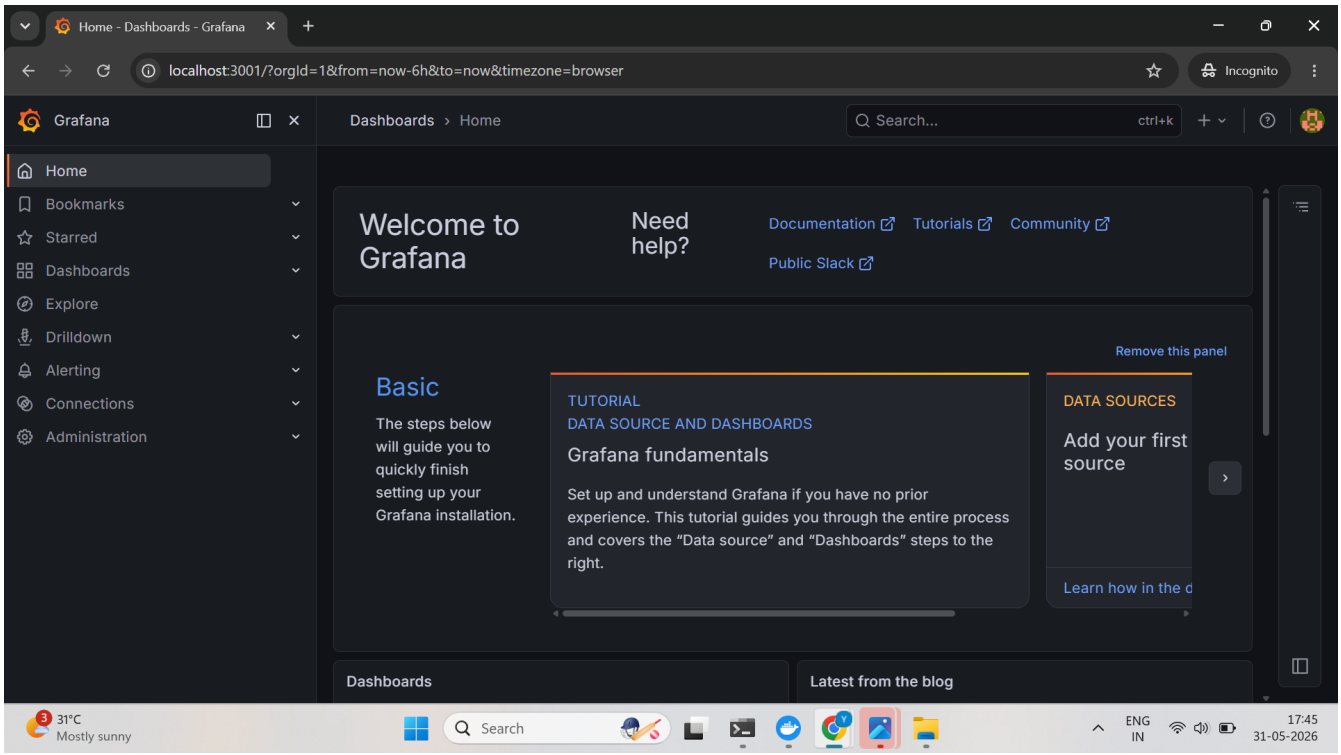
Promtail Configuration Example

server: http_listen_port: 9080 clients: - url: http://loki:3100/loki/api/v1/push scrape_configs: - job_name: nginx static_configs: - targets: - localhost

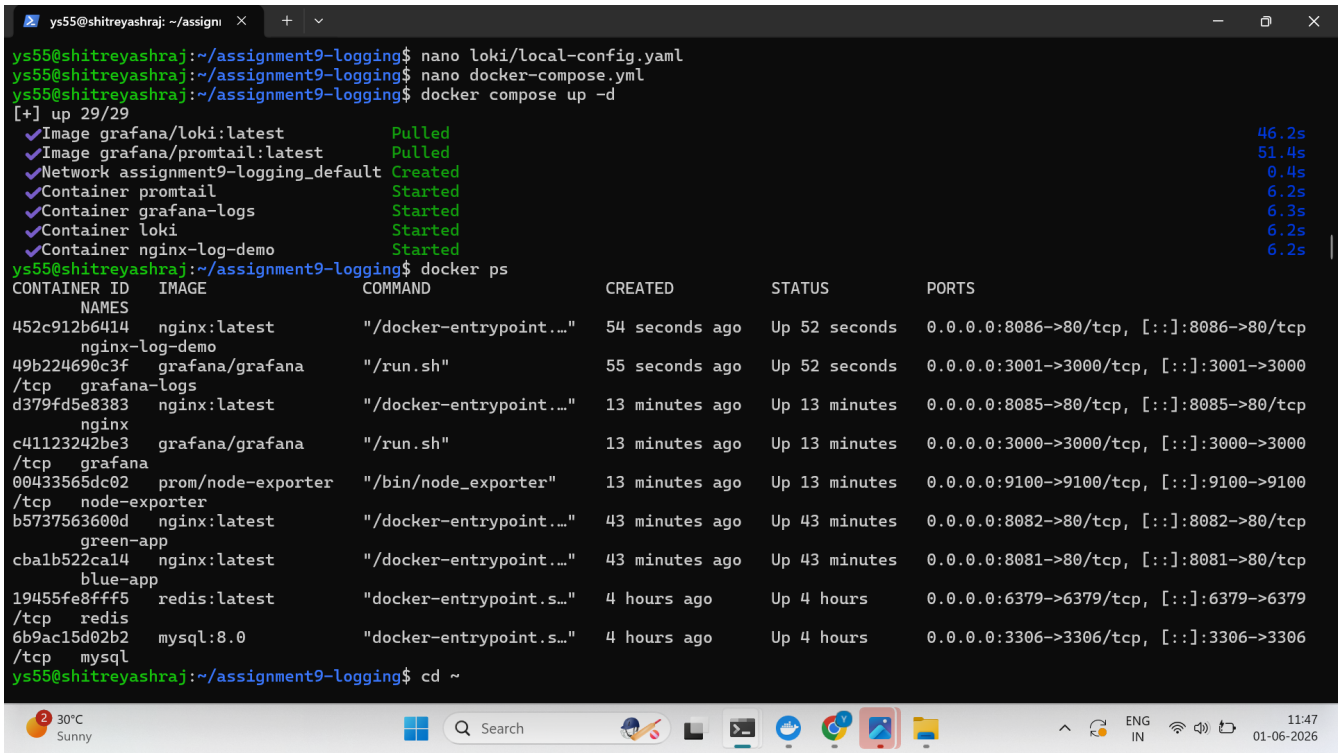
Log Query Examples

{job="nginx"} {container="nginx-log-demo"} {job="docker"} |= "error" {job="docker"} |= "warning"

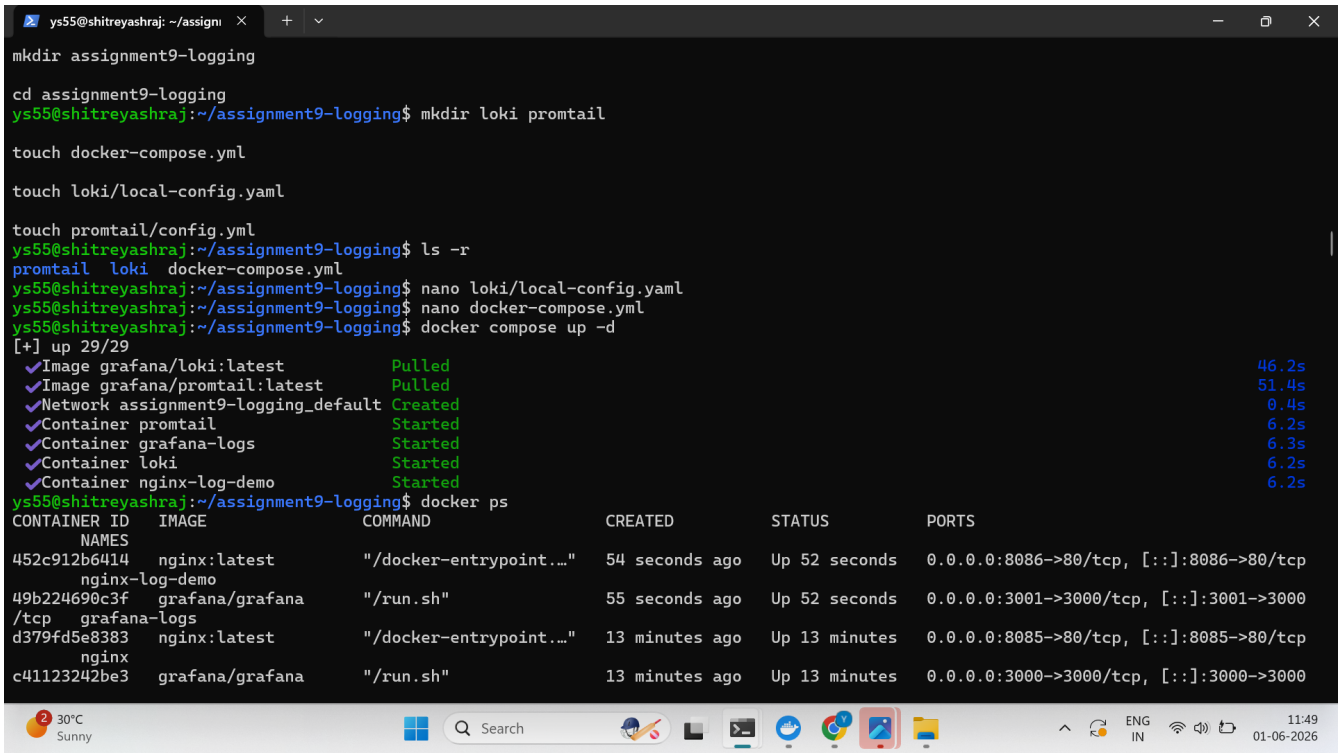
Screenshot Evidence



Screenshot Evidence



Screenshot Evidence



Dashboard Screenshots & Testing Results

✓ Loki container deployed successfully.
✓ Promtail container collecting logs.
✓ Grafana dashboard accessible.
✓ Application logs collected centrally.
✓ Nginx logs collected successfully.
✓ Log filtering and querying validated.
✓ Docker containers operational.

Troubleshooting Guide

Issue: Logs not appearing in Grafana.
Resolution: Verify Promtail configuration and Loki connectivity.

Issue: Loki service unavailable.
Resolution: Check Docker container status and logs.

Issue: Dashboard shows no data.
Resolution: Verify Grafana datasource configuration.

Rollback / Recovery Procedure

1. Stop logging stack containers. 2. Restore previous Loki and Promtail configurations. 3. Redeploy Docker Compose stack. 4. Verify log collection functionality.

Key Decisions and Configuration Explanation

Loki was selected because it is lightweight and integrates natively with Grafana. Promtail simplifies log collection from containers and applications. Grafana provides centralized visibility, search, filtering, and dashboard capabilities. Centralized logging improves troubleshooting, auditing, and operational monitoring.